

1. Determine the dc bias voltage V_{CE} , the current I_C and the ac emitter resistance r_e for the voltage-divider configuration given in Fig. 1. Let $V_{CC} = +20V$, $\beta = 100$, $R_1 = 42k\Omega$, $R_2 = 4.2k\Omega$, $R_C = 12k\Omega$, $R_E = 1.5k\Omega$.

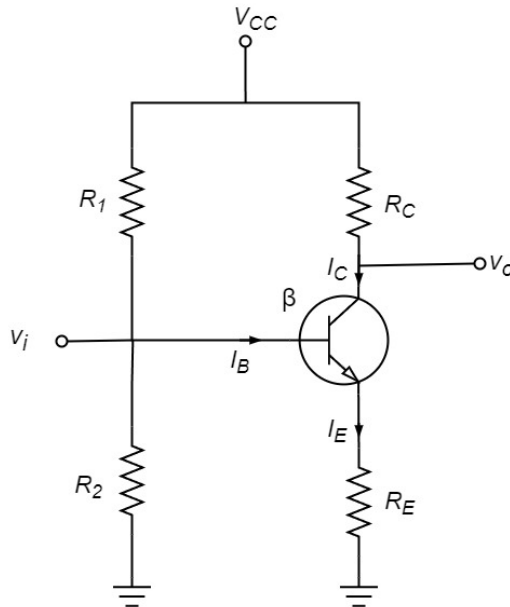


Figure 1

Solution

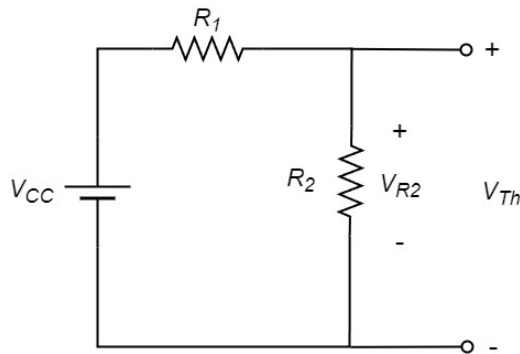


Figure 2

Replacing Fig. 2 with an equivalent Thevenin circuit, we get

$$\begin{aligned}
 R_{Th} &= R_1 || R_2 \\
 &= \frac{(42k\Omega)(4.2k\Omega)}{42k\Omega + 4.2k\Omega} \\
 &= 3.82k\Omega
 \end{aligned}$$

The Thevenin voltage is given by

$$\begin{aligned} V_{Th} &= \frac{R_2 V_{CC}}{R_1 + R_2} \\ &= \frac{(4.2k\Omega)(20V)}{42k\Omega + 4.2k\Omega} \\ &= 1.82V \end{aligned}$$

The Thevenin network is redrawn as in Fig.3

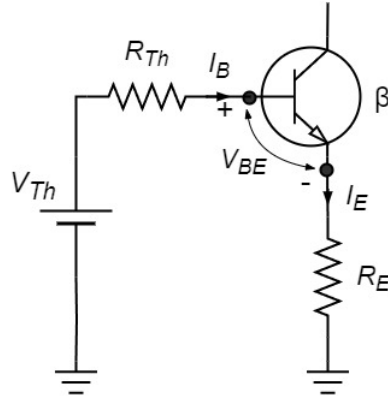


Figure 3

Similarly,

$$\begin{aligned} I_B &= \frac{V_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\ &= \frac{1.82V - 0.7V}{3.82k\Omega + (101)(1.5k\Omega)} \\ &= \frac{1.12V}{3.82k\Omega + 151.5k\Omega} \\ &= 7.21\mu A \end{aligned}$$

$$\begin{aligned} I_C &= \beta I_B \\ &= (100)(7.21\mu A) \\ &= 0.72mA \end{aligned}$$

$$\begin{aligned} I_E &= (\beta + 1)I_B \\ &= (101)(7.21\mu A) \\ &= 0.727mA \end{aligned}$$

The dc bias voltage

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C - I_E R_E \\ &= 20V - (0.72mA)(12k\Omega) - (0.727mA)(1.5k\Omega) \\ &= 20V - 8.64V - 1.09V \\ &= 10.26V \end{aligned}$$

The ac emitter resistance r_e with the thermal voltage $V_T = 26mV$

$$\begin{aligned}
 r_e &= \frac{V_T}{I_E} \\
 &= \frac{26 \text{ mV}}{I_E} \\
 &= \frac{26 \text{ mV}}{0.727 \text{ mA}} \\
 &= 35.76 \Omega
 \end{aligned}$$

2. Determine the value of I_C , V_{CE} and ac emitter resistance r_e for the network of Fig. 4. Let $V_{CC} = +12\text{V}$, $\beta = 100$, $R_B = 250\text{k}\Omega$, $R_C = 5\text{k}\Omega$, $R_E = 1.2\text{k}\Omega$.

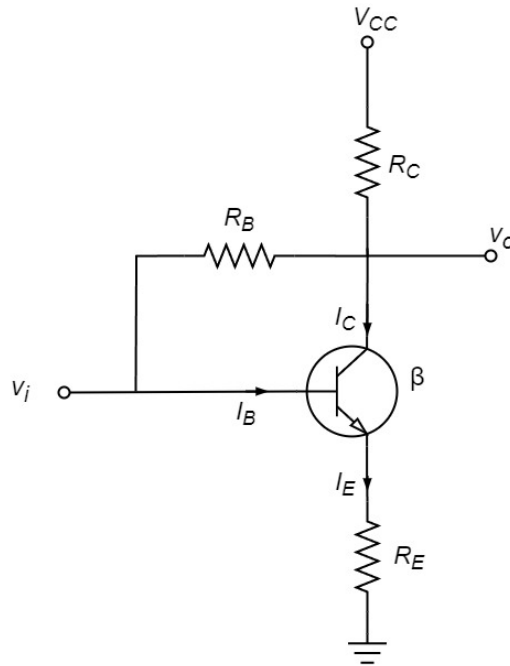


Figure 4

Solution

Writing the Kirchoff's voltage law, we get

$$V_{CC} - V_{BE} - (\beta + 1)I_B R_C - (\beta + 1)I_B R_E - I_B R_B = 0$$

Therefore, solving for I_B , we get

$$\begin{aligned}
 I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_C + (\beta + 1)R_E} \\
 &= \frac{12\text{V} - 0.7\text{V}}{250\text{k}\Omega + (101)(5\text{k}\Omega + 1.2\text{k}\Omega)} \\
 &= 12.89\mu\text{A}
 \end{aligned}$$

$$\begin{aligned}
 I_C &= \beta I_B \\
 &= (100)(12.89\mu\text{A}) \\
 &= 1.29\text{mA}
 \end{aligned}$$

$$\begin{aligned}
 I_E &= (\beta + 1)I_B \\
 &= (101)(12.89\mu\text{A}) \\
 &= 1.30\text{mA}
 \end{aligned}$$

$$\begin{aligned}
 V_{CE} &= V_{CC} - I_E R_C - I_E R_E \\
 &= 12 \text{ V} - (1.30 \text{ mA})(5 \text{ k}\Omega) - (1.30)(1.2 \text{ k}\Omega) \\
 &= 12 \text{ V} - 8.06 \text{ V} \\
 &= 3.94 \text{ V}
 \end{aligned}$$

The ac emitter resistance r_e with the thermal voltage $V_T = 26 \text{ mV}$

$$\begin{aligned}
 r_e &= \frac{V_T}{I_E} \\
 &= \frac{26 \text{ mV}}{I_E} \\
 &= \frac{26 \text{ mV}}{1.30 \text{ mA}} \\
 &= 20 \Omega
 \end{aligned}$$

3. Refer to the circuit in the figure below. Determine the total average power, reactive power, and complex power at the source and at the load.

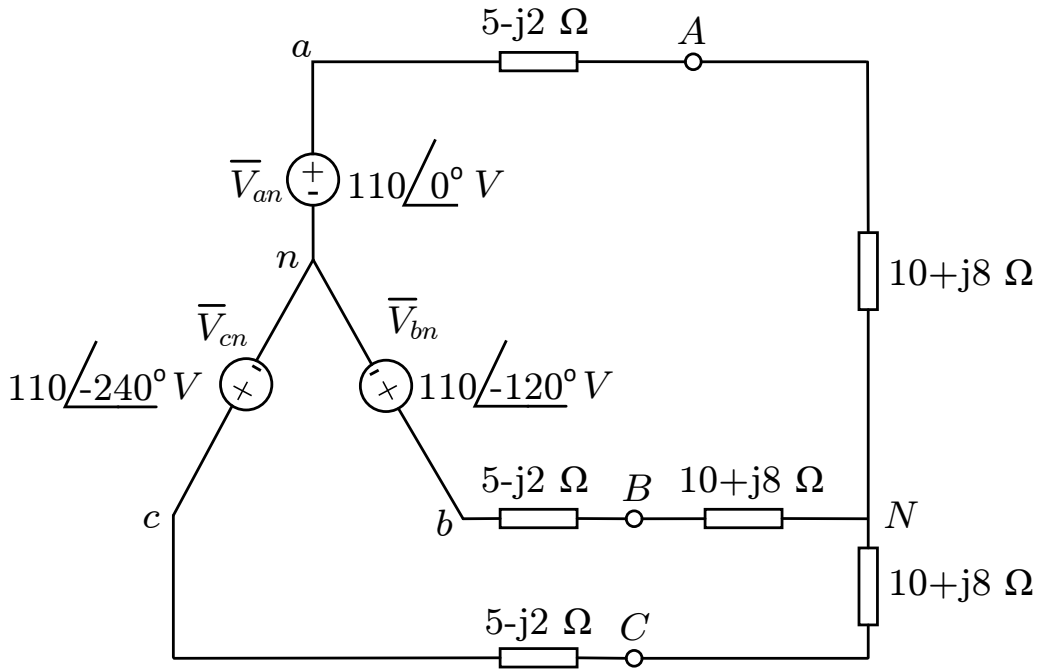


Figure 5 Three-wire Y-Y system.

Solution:

It is sufficient to consider one phase, as the system is balanced. For phase a ,

$$\bar{V}_p = 110\angle 0^\circ \text{ V and } \bar{I}_p = 6.81\angle -21.8^\circ \text{ A}$$

Thus, at the source, the complex power is

$$\begin{aligned}
 \bar{S}_s &= -3\bar{V}_p \bar{I}_p^* = -3(110\angle 0^\circ)(6.81\angle 21.8^\circ) \\
 &= -2247\angle 21.8^\circ = -(2087 + j834.6) \text{ VA}
 \end{aligned}$$

The real or average power is -2087 W and the reactive power is -834.6 VAR .

At the load, the complex power is

$$\bar{S}_L = 3|\bar{I}_p|^2 \bar{Z}_p$$

where, $\bar{Z}_p = 10 + j8 = 12.81\angle 38.66^\circ$ and $\bar{I}_p = \bar{I}_a = 6.81\angle -21.8^\circ$.

Hence,

$$\bar{S}_L = 3(6.81)^2 12.81 / 38.66^\circ = 1782 / 38.66^\circ = (1392 + j1113) \text{ VA}$$

The real power is 1391.7 W, and the reactive power is 1113.3 VAR.

The difference between the two complex powers is absorbed by the line impedance $(5 - j2) \Omega$. To show that this is the case, we find the complex power of the line as

$$\bar{S}_l = 3|\bar{I}_p|^2 \bar{Z}_l = 3(6.81)^2 (5 - j2) = 695.6 - j278.3 \text{ VA}$$

which is the difference between $\bar{S}_s$ and $\bar{S}_L$; that is, $\bar{S}_s + \bar{S}_l + \bar{S}_L = 0$, as expected.

4. Two balanced loads are connected to a 240 kV, 60 Hz line, as shown in Figure below. Load 1 draws 30 kW at a power factor of 0.6 lagging, while load 2 draws 45 kVAR at a power factor of 0.8 lagging. Assuming the abc sequence, determine:

- (A) the complex, real, and reactive powers of the combined load
- (B) the line currents
- (C) the kVAR rating of the three capacitors Δ -connected in parallel with the load that will raise the power factor to 0.9 lagging and the capacitance of each capacitor.

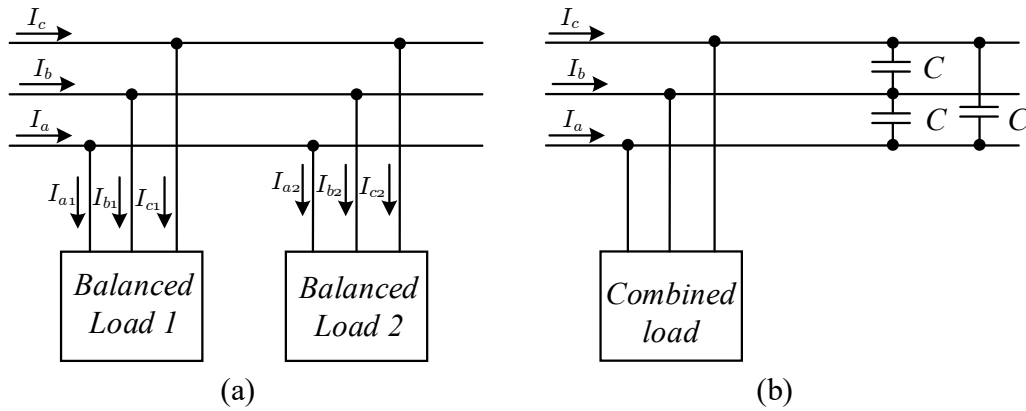


Figure 6 (a) The originally balanced loads, (b) the combined load with improved power factor.

Solution:

Part A

For load 1, given that $P_1 = 30 \text{ kW}$ and $\cos \theta_1 = 0.6$, then $\sin \theta_1 = 0.8$. Hence,

$$S_1 = |\bar{S}_1| = \frac{P_1}{\cos \theta_1} = \frac{30 \text{ kW}}{0.6} = 50 \text{ kVA}$$

and $Q_1 = S_1 \sin \theta_1 = 50(0.8) = 40 \text{ kVAR}$. Thus, the complex power due to load 1 is

$$\bar{S}_1 = P_1 + jQ_1 = 30 + j40 \text{ kVA}$$

For load 2, if $Q_2 = 45 \text{ kVAR}$ and $\cos \theta_2 = 0.8$, then $\sin \theta_2 = 0.6$. So,

$$S_2 = |\bar{S}_2| = \frac{Q_2}{\sin \theta_2} = \frac{45 \text{ kVAR}}{0.6} = 75 \text{ kVA}$$

and $P_2 = S_2 \cos \theta_2 = 75(0.8) = 60 \text{ kW}$. Therefore, the complex power due to load 2 is

$$\bar{S}_2 = P_2 + jQ_2 = 60 + j45 \text{ kVA}$$

From the above two equations, the total complex power due to the load is

$$\bar{S} = \bar{S}_1 + \bar{S}_2 = 90 + j85 \text{ kVA} = 123.8 / 43.36^\circ \text{ kVA}$$

which has a power factor of $\cos(43.36)^\circ = 0.727$ lagging. The real power is 90 kW, while the reactive power is 85 kVAR.

Part B

Since $S = \sqrt{3}V_L I_L$, the line current is

$$I_L = \frac{S}{\sqrt{3}V_L}$$

We apply this to each load, keeping in mind that for both loads, $V_L = 240 \text{ kV}$.
For load 1,

$$I_{L1} = \frac{50000}{\sqrt{3} 240000} = 120.28 \text{ mA}$$

Since the power factor is lagging, the line current lags the line voltage by $\theta_1 = \cos^{-1}0.6 = 53.13^\circ$. Thus,

$$\bar{I}_{a1} = I_{L1}/\underline{\theta_1} = 120.28/\underline{-53.13^\circ} \text{ mA}$$

For load 2,

$$I_{L2} = \frac{75000}{\sqrt{3} 240000} = 180.42 \text{ mA}$$

and the line current lags the line voltage by $\theta_2 = \cos^{-1}0.8 = 36.87^\circ$. Hence,

$$\bar{I}_{a2} = I_{L2}/\underline{\theta_2} = 180.42/\underline{-36.87^\circ} \text{ mA}$$

The total line current is

$$\begin{aligned}\bar{I}_a &= \bar{I}_{a1} + \bar{I}_{a2} = 120.28/\underline{-53.13^\circ} + 180.42/\underline{-36.87^\circ} \\ &= (72.168 - j96.224) + (144.336 - j108.252) \\ &= 216.5 - j204.472 = 297.82/\underline{-43.36^\circ} \text{ mA}\end{aligned}$$

Alternatively, we could obtain the current from the total complex power as

$$\begin{aligned}I_L &= \frac{123800}{\sqrt{3} 240000} = 297.82 \text{ mA} \\ \bar{I}_a &= I_L/\underline{\theta} = 297.82/\underline{-43.36^\circ} \text{ mA}\end{aligned}$$

which is the same as before. The other line currents, $\bar{I}_b$ and $\bar{I}_c$, can be obtained according to the abc sequence (i.e., $\bar{I}_b = 297.82/\underline{-163.36^\circ} \text{ mA}$ and $\bar{I}_c = 297.82/\underline{76.64^\circ} \text{ mA}$).

Part C

We can find the reactive power needed to bring the power factor to 0.9 lagging.

$$Q_C = P(\tan\theta_{old} - \tan\theta_{new})$$

where $P = 90 \text{ kW}$, $\theta_{old} = 43.36^\circ$, and $\theta_{new} = \cos^{-1}0.9 = 25.84^\circ$.
Hence,

$$Q_C = 90000(\tan 43.36^\circ - \tan 25.84^\circ) = 41.4 \text{ kVAR}$$

This reactive power is for the three capacitors. For each capacitor, the rating $Q'_c = \frac{Q_C}{3} = 13.8 \text{ kVAR}$. The required capacitance is

$$C = \frac{Q'_c}{\omega V_{rms}^2}$$

Since the capacitors are Δ connected as shown in the figure, V_{rms} in the above formula is the line-to-line or line voltage, which is 240 kV . Thus,

$$C = \frac{13800}{(2\pi 60)(240000)^2} = 635.5 \text{ pF}$$